ALEX MOREHEAD

Machine Learning & Computational Biology Researcher

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"Machine Learning & Computational Biology Researcher with over 6 years of experience developing cutting-edge deep learning algorithms for geometric (e.g., structural biology) data. Track record of bringing new ideas to life and implementing them using AI-standard programming languages/frameworks. Experienced leading research teams, fostering collaborations, and presenting research findings at top conferences."

SKILLS

Research | Teaching | Python | PyTorch | PyG | PyTorch Lightning | W&B | Hydra | Slurm | Git | LaTeX

Geometric deep learning | Generative modeling | Graph neural networks | AI4Science | CompBio/Chem

EXPERIENCE

Hopper Postdoctoral Fellow (Supervisor: Dr. Michael Mahoney)

Lawrence Berkeley National Laboratory | International Computer Science Institute

Jun 2025 – Ongoing | California, USA

Awarded Berkeley Lab's **2025 Hopper Postdoctoral Fellowship** in computing sciences, courtesy of NERSC.

- Research and develop **scientific foundation models**, including the **Zatom** series of models, at LBNL and ICSI.
- Conduct **large-scale** deep learning HPC experiments and make contributions to **open-source code** along the way.

Deep learning | Generative models | AI4Science | CompBio/Chem | Scientific machine learning

Graduate Research Assistant (Supervisor: Dr. Jianlin Cheng)

University of Missouri | Bioinformatics & Machine Learning Lab

Aug 2020 – May 2025 | Missouri, USA

Awarded two competitive first-year PhD fellowships and the **outstanding PhD** awards for EECS/CS graduate studies.

- Researched geometric and generative modeling methods for bioinformatics, yielding **20+ academic works**.
- Developed two state-of-the-art protein representation learning methods along with the first diffusion model to successfully generate valid large 3D molecules and flow model for blind protein-ligand docking. Published results at top-tier machine learning conferences (**ICLR**) and scientific venues (**ISMB** / **Bioinformatics**, **CASP16**, **Nature Machine Intelligence**, & **Communications Chemistry**).
- Introduced the first deep learning benchmark for **broadly applicable** protein-ligand docking.
- Orchestrated several **large-scale** deep learning HPC experiments and contributed to **open-source code**.

Deep learning | Geometric & graph representation learning | Generative models | Structural bioinformatics

Research Intern (Supervisor: Dr. Jeffrey Ruffolo)

Profluent Bio | AI Research Lab

May 2023 – Aug 2023 | California, USA

- Created **MMDiff**, the first SE(3) diffusion model for joint sequence-structure generation of DNA, RNA, and proteins, which achieved a **9% nucleic acid design success rate**. Presented at NeurIPS 2023 MLSB. **Paper** and **Code**.

Diffusion models | Structure generation | Sequence generation | Differential geometry | Google Cloud

Teaching Assistant (Supervisor: Lukin Murphy)

EnCircle Technologies | Programming Curriculum

May 2022 – Aug 2023 | Missouri, USA

- Crafted and taught special education computer science curriculum for neurodivergent youth, ultimately implementing **reinforcement learning algorithms in a collaborative Minecraft environment**. **Open-source curriculum**.

Teaching | Curriculum development | Computer science | Python programming | Machine learning

ACHIEVEMENTS

Berkeley Lab's 2025 Hopper Postdoctoral Fellow
• Won a *competitive* postdoctoral fellowship for recent PhD graduates in computing sciences.

LoG Top-10 Reviewer
• Awarded large monetary prize for being a *top-3%* reviewer for the 2023 Learning on Graphs (LoG) conference.

Dean's Engineering Excellence & O'Neill Graduate Fellowships and Top-Ranked EECS/CS PhD Student Awards
• Won *two* competitive graduate fellowships for first-year PhD students and the *outstanding EECS & CS PhD student* awards.

Public outreach
• Research featured in *two* public-facing venues including HPCwire and Marktechpost.

INVITED TALKS

- Zatom-1: Towards a Multimodal Foundation Model for 3D Molecules and Materials, DL4Sci @ UCB, July 2026, Berkeley, CA, USA
- Zatom-1: Towards a Multimodal Foundation Model for 3D Molecules and Materials, LeMaterial Reading Group, March 2026, Remote
- FlowDock: Geometric Flow Matching for Protein-Ligand Docking and Affinity Prediction, 3DSIG @ ISMB, Jul 2025, Liverpool, UK
- Geometric Deep Learning and Generative Models For Protein-Ligand Docking, Computational modeling of molecular structures course @ MU, Nov 2024, Columbia, MO, USA
- RNA-FrameFlow for *de novo* 3D RNA backbone design, SPIGM @ ICML, Jul 2024, Vienna, AT
- A Gated Graph Transformer for Protein Complex Structure Quality Assessment, 3DSIG @ ISMB, Jul 2023, Lyon, FR
- Geometry-Complete Perceptron Networks for 3D Molecular Graphs, AI2ASE @ AAAI, Feb 2023, Washington D.C., USA
- Neural Diffusion Models: Next-Generation Generative Deep Learning, Advanced topics in deep learning course @ MU, Nov 2022, Columbia, MO, USA
- Geometric Transformers for Protein Interface Contact Prediction, Xuefeng Cui Lab @ Shandong University, May 2022, Remote

EDUCATION

M.S. & Ph.D. in Computer Science

University of Missouri | O'Neill and College of Engineering Dean's Graduate Fellow

Aug 2020 - May 2025

PhD Topic:
- Geometric Deep Learning & Generative Modeling of 3D Biomolecules

Collaborations:
- University of Cambridge, Research Labs of Prof. Pietro Lió & Prof. Sir Tom Blundell

Coursework:
- Highest distinction (GPA=4.0/4.0)

B.S. in Computer Science

Missouri Western State University | General Studies and Outstanding Graduate Honors

Aug 2016 - May 2020

- Graduated **top of class** among all 2020 graduates in computer science, mathematics, and physics (GPA=4.0/4.0)

REFERENCES

Dr. Michael Mahoney
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Dr. Wahid Bhimji
@ Lawrence Berkeley National Laboratory
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Dr. Jianlin Cheng
@ University of Missouri
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Dr. Dong Xu
@ University of Missouri
xudong@missouri.edu

PUBLICATIONS

Conference & Workshop Proceedings

[1] H. La, A. Gupta, **A. Morehead**, J. Cheng, and M. Zhang, "MegaFold: System-Level Optimizations for Accelerating Protein Structure Prediction Models," in *ISC High Performance*, 2026.

[2] **A. Morehead*** et al., "Zatom-1: Towards a Multimodal Foundation Model for 3D Molecules and Materials," in *ICLR Workshop on Foundation Models for Science*, under review with an ML venue, 2026.

[3] H. B. Ta, **A. Morehead**, N. B. Erichson, N. Fioretto, and S. Madireddy, "Choosing the Right Representations: Task-Aware Fusion for Multimodal Materials Language Models," in *NeurIPS AI for Accelerated Materials Design (AI4Mat) Workshop*, 2026.

[4] A. A. Elhag*, A. Raja*, **A. Morehead***, S. M. Blau, G. M. Morris, and M. M. Bronstein, "Learning Inter-Atomic Potentials without Explicit Equivariance," in *NeurIPS Machine Learning in Structural Biology (MLSB) Workshop*, under review with an ML venue, 2025.

[5] C. K. Joshi et al., "gRNAde: Geometric Deep Learning for 3D RNA Inverse Design," in *Thirteenth International Conference on Learning Representations (ICLR)*, selected as an ICLR 2025 *spotlight*, with follow-up work achieving wet-lab validation, 2025.

[6] **A. Morehead** and J. Cheng, "FlowDock: Geometric Flow Matching for Generative Protein-Ligand Docking and Affinity Prediction," in *Intelligent Systems for Molecular Biology (ISMB)*, presented as a CASP16 top-5 method, 2025.

[7] M. Pourmirzaei, **A. Morehead**, F. Esmaili, J. Ren, M. Pourmirzaei, and D. Xu, "GCP-VQVAE: A Geometry-Complete Language for Protein 3D Structure," in *NeurIPS AI4Science Workshop*, under review with an ML venue, 2025.

[8] A. R. Jamasb*, **A. Morehead***, C. K. Joshi*, Z. Zhang*, et al., "Evaluating Representation Learning on the Protein Structure Universe," in *Twelfth International Conference on Learning Representations (ICLR)*, also at NeurIPS 2023 MLSB (poster), 2024.

[9] X. Chen*, **A. Morehead***, J. Liu, and J. Cheng, "A gated graph transformer for protein complex structure quality assessment and its performance in CASP15," in *Intelligent Systems for Molecular Biology (ISMB)*, 2023.

[10] **A. Morehead**, A. Bhatnagar, J. A. Ruffolo, and A. Madani, "Towards Joint Sequence-Structure Generation of Nucleic Acid and Protein Complexes," in *NeurIPS Machine Learning in Structural Biology (MLSB) Workshop*, 2023.

[11] **A. Morehead***, W. Chantapakul*, and J. Cheng, "Semi-Supervised Graph Learning Meets Dimensionality Reduction," in *IEEE International Conference on Machine Learning and Applications*, 2023.

[12] E. Soltanikazemi*, R. S. Roy*, F. Quadir, N. Giri, **A. Morehead**, and J. Cheng, "DRLComplex: Reconstruction of protein quaternary structures using deep reinforcement learning," in *International Conference on Intelligent Biology and Medicine*, 2023.

[13] **A. Morehead**, C. Chen, and J. Cheng, "Geometric Transformers for Protein Interface Contact Prediction," in *Tenth International Conference on Learning Representations (ICLR)*, 2022.

[14] M. Shoman*, A. Aboah*, **A. Morehead**, Y. Duan, A. Daud, and Y. Adu-Gyamfi, "A Region-Based Deep Learning Approach to Automated Retail Checkout," in *Proceedings of the IEEE/CVF CVPR Workshops*, 2022.

[15] M. Gao et al., "High-Performance Deep Learning Toolbox for Genome-Scale Prediction of Protein Structure and Function," in *IEEE/ACM Machine Learning with Graphs in High Performance Computing Environments (MLHPC) Workshop*, 2021.

[16] **A. Morehead**, L. Ogden, G. Magee, R. Hosler, B. White, and G. Mohler, "Low Cost Gunshot Detection using Deep Learning on the Raspberry Pi," in *IEEE International Conference on Big Data*, 2019.

Journal Articles

[17] **A. Morehead***, J. Liu*, P. Neupane*, and J. Cheng, "Artificial intelligence methods for protein structure and interaction prediction: Recent advances and challenges," *Current Opinion in Structural Biology*, 2026, invited issue on AI Methodologies in Structural Biology.

[18] **A. Morehead*** et al., "Flow matching for generative modeling in bioinformatics and computational biology," *Nature Machine Intelligence*, 2026.

[19] R. Anand*, C. K. Joshi*, **A. Morehead**, et al., "RNA-FrameFlow for de novo 3D RNA Backbone Design," *TMLR*, 2025, presented at ICML 2024 SPIGM (*oral*) and AI4Science (*spotlight*) workshops.

[20] **A. Morehead**, "Geometric deep learning and generative modeling of 3D biomolecules," *University of Missouri-Columbia*, 2025, Ph.D. dissertation.

[21] **A. Morehead**, N. Giri, J. Liu, P. Neupane, and J. Cheng, "Assessing the potential of deep learning for protein-ligand docking," *Nature Machine Intelligence*, 2025, also presented as an ICML 2024 AI4Science *spotlight*, top 20% - 30/159.

[22] **A. Morehead**, J. Liu, P. Neupane, N. Giri, and J. Cheng, "Protein-ligand structure and affinity prediction in CASP16 using a geometric deep learning ensemble and flow matching," *Proteins: Structure, Function, and Bioinformatics*, 2025, presented as a CASP16 top-5 method.

[23] **A. Morehead** and J. Cheng, "Geometry-complete diffusion for 3D molecule generation and optimization," *Communications Chemistry*, 2024, in Nature Communications subjournal, also presented at ICLR 2023 MLDD (poster).

[24] **A. Morehead** and J. Cheng, "Geometry-complete perceptron networks for 3D molecular graphs," *Bioinformatics*, 2024, also presented at AAAI 2023 DLG (poster) and AI2ASE (*oral*).

[25] C. Chen, X. Chen, **A. Morehead**, T. Wu, and J. Cheng, "3D-equivariant graph neural networks for protein model quality assessment," *Bioinformatics*, 2023.

[26] M. F. Lensink et al., "Impact of AlphaFold on structure prediction of protein complexes: The CASP15-CAPRI experiment," *Proteins: Structure, Function, and Bioinformatics*, 2023.

[27] **A. Morehead**, C. Chen, A. Sedova, and J. Cheng, "DIPS-Plus: The enhanced database of interacting protein structures for interface prediction," *Scientific Data*, 2023.

[28] **A. Morehead** and J. Cheng, "Protein Structure Accuracy Estimation using Geometry-Complete Perceptron Networks," *Protein Science*, 2023.

[29] O. Kouckya et al., "Synthetic Biology Bicistronic Designs Support Gene Expression Equally Well in vitro and in vivo," *AJUR*, 2020.

Preprints

[30] M. Cretu* et al., *Zatom-2: Multitask Pretraining on Atomistic Data for Generative Modeling across Domains*, on arXiv and under review with an ML venue, serving as the senior author and project co-lead, 2026.

[31] Z. Wang, A. Jamasb, M. Hajij, **A. Morehead**, L. Braithwaite, and P. Liò, *Topotein: Topological Deep Learning for Protein Representation Learning*, arXiv, 2025.

[32] S. Mahmud, **A. Morehead**, and J. Cheng, *Accurate prediction of protein tertiary structural changes induced by single-site mutations with equivariant graph neural networks*, bioRxiv, 2023.

[33] A. Shanehsazzadeh et al., *Unlocking de novo antibody design with generative artificial intelligence*, bioRxiv, demonstrating wet-lab validation and presenting follow-up work at NeurIPS 2023 MLSB (poster), 2023.

[34] **A. Morehead**, X. Chen, T. Wu, J. Liu, and J. Cheng, *EGR: Equivariant Graph Refinement and Assessment of 3D Protein Complex Structures*, arXiv, 2022.

FUNDING

- Hopper Postdoctoral **Fellowship**, U.S. Department of Energy, ~$300,000
- Dean's Engineering Excellence **Fellowship**, University of Missouri, $10,000
- O'Neill Graduate **Fellowship**, University of Missouri, $10,000
- XSEDE Hardware **Grant**, U.S. National Science Foundation, $500
- Top-10 Reviewer **Award**, Learning on Graphs Conference, $500

SERVICE

- **Reviewing**:
  - **Conferences & Workshops**:
    - Neural Information Processing Systems (NeurIPS) **[2021-…]**
      - AI4Sci, AI4D3, GenBio, & MLSB workshops **[2021-2025]**
    - International Conference on Learning Representations (ICLR) **[2024-…]**
      - GEM, AI4NA, and FPI workshops **[2024-2025]**
    - International Conference on Machine Learning (ICML) **[2024-…]**
      - AI4Sci, CB, AccMLBio, & SPIGM workshops **[2024-2025]**
      - Served on the ICML ML4LMS workshop's organizing committee **[2024]**
    - Association for the Advancement of Artificial Intelligence (AAAI) **[2025-2026]**
    - Learning on Graphs Conference (LoG) of PMLR **[2024-2025]**
      - Served on the LoG conference's organizing committee **[2024-2026]**
    - ACM Conference on Bioinformatics, Computational Biology, & Health Informatics (ACM BCB) **[2024-…]**
  - **Journals**:
    - Transactions on Machine Learning Research (TMLR) **[2025-…]**
    - Nature Machine Intelligence & Nature Methods **[2023-…]**, Nature Communications **[2025-…]**
    - Bioinformatics, Briefings in Bioinformatics, BMC Bioinformatics, JCIM, and CSB Journal **[2024-…]**
    - IEEE Trans. on Neural Nets. & Learning Systems (TNNLS) & Emerging Topics in Comp. Intell. (TETCI) **[2023-…]**

- **Mentoring**:
  - **Undergraduate Students**:
    - Rishabh Anand (NUS, CS major), now a PhD student at MIT
    - Kyler Zook (MU, CS major)
    - Gautam Bhooma (UCB, CS-ChemE dual major)
    - Sai Srikar Tummala (UCB, data science major)
    - Salman Ibrahim Koshy (UCB, operations research and data science major)

- Sarvagya Goyal (UCB, CS and data science major)
- Nathan Sekar (Rutgers, CS major)

- **Graduate Students**:
  - Miruna Cretu (PhD student at Cambridge), now at Boltz
  - Antonia Panescu (PhD student at Yale)
  - Tynan Perez (PhD student at MIT)
  - Ahmed A Elhag (PhD student at Oxford)
  - Arun Raja (PhD student at Oxford)
  - Mahdi Pourmirzaei (PhD student at MU)
  - Zhiyu Wang (Master's student at Cambridge), now a Research Scientist at Sapient Intelligence
  - Rylie Weaver (Master's student at CGU), now a PhD student at UTK
  - Pushpa Kumar Balan (Master's student at UCMO)